NATIONAL UNIVERSITY OF SINGAPORE

IT5005 – ARTIFICIAL INTELLIGENCE

(Semester 2: AY2024/25)

Final Assessment

Time Allowed: 2 Hours

INSTRUCTIONS

STUDENT NUMBER

A

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1. Write your **Student Number** on the right AND, using pen or pencil, shade the corresponding circle **completely** in the grid for each digit or letter. DO NOT WRITE YOUR NAME!
2. Zero mark will be given if you write/shade your Student Number incompletely or incorrectly.
3. Write your Student Number at the top of page 3.
4. This answer sheet comprises **TWELVE (12) pages**.
5. All questions must be answered in the space provided; no extra sheets will be accepted as answers.
6. You must submit only this **ANSWER SHEET** and no other documents.
7. An excerpt of the question may be provided to aid you in answering in the correct box. It is not the exact question. You should still refer to the original question in the question paper.
8. You may write your answers using pencil (at least 2B) or pen as long as it is legible (no red ink, please).
9. The maximum mark for this paper is 100.
10. **Marks may be deducted** for (i) illegible handwriting, and/or (ii) excessively long explanations.

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Question	**Marks**	**Remarks**
Q1	/ 15	
Q2	/ 15	
Q3	/ 20	
Q4	/ 20	
Q5	/ 20	
Q6	/ 10	
Total	**/ 100**	

1. **Markov Chains.** [15 marks]

a. Missing Values [4 marks]

x_1	x_2	x_3	x_4
0	0	0.2	0.1

b. Probability distribution one hour from now [6 marks]

Given $\boldsymbol{P}(R_t) = \begin{bmatrix} P(R_t = A) \\ P(R_t = C) \\ P(R_t = B) \\ P(R_t = O) \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$, the query is $\boldsymbol{P}(R_{t+1})$.

$$\boldsymbol{P}(R_{t+1}) = \alpha \begin{bmatrix} 0.5 & 0.4 & 0.1 & 0 \\ 0.3 & 0.6 & 0.1 & 0 \\ 0 & 0.6 & 0.2 & 0.2 \\ 0.4 & 0.3 & 0.2 & 0.1 \end{bmatrix}^T \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0.4 \\ 0.1 \\ 0 \end{bmatrix}$$

c. Probability distribution for the last hour of the shift [5 marks]

Given $\boldsymbol{P}(R_1) = \begin{bmatrix} P(R_1 = A) \\ P(R_1 = C) \\ P(R_1 = B) \\ P(R_1 = O) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$, the query is $\boldsymbol{P}(R_4)$.

$$\boldsymbol{P}(R_4) = \alpha \left(\begin{bmatrix} 0.5 & 0.4 & 0.1 & 0 \\ 0.3 & 0.6 & 0.1 & 0 \\ 0 & 0.6 & 0.2 & 0.2 \\ 0.4 & 0.3 & 0.2 & 0.1 \end{bmatrix}^T \right)^3 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

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2. **Hidden Markov Models.** [15 marks]

(a) Prior distribution and transition probability matrix [2 marks]

Prior distribution	Transition probability
$\boldsymbol{P}(X_0) = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$	$T = \begin{bmatrix} 0.4 & 0.6 \\ 0.7 & 0.36 \end{bmatrix}$

(b) Probability that first bit is received correctly. [5 marks]

The Query: $P(X_1 = -1 | E_1 = -1)$

$$P(X_1 = -1 | E_1 = -1) = \frac{P(X_1 = -1, E_1 = -1)}{P(E_1 = -1)}$$

$$= \frac{P(E_1 = -1 | X_1 = -1) P(X_1 = -1)}{P(E_1 = -1)}$$

$$P(E_1 = -1) = P(E_1 = -1 | X_1 = -1) P(X_1 = -1) + P(E_1 = -1 | X_1 = +1) P(X_1 = +1)$$

$$= 0.8 * 0.5 + 0.2 * 0.5 = 0.5$$

$$P(X_1 = -1 | E_1 = -1) = \frac{P(E_1 = -1 | X_1 = -1) P(X_1 = -1)}{P(E_1 = -1)}$$

$$= \frac{0.8 * 0.5}{0.5}$$

$$= 0.8$$

(c) Probability that second bit is also received correctly. [8 marks]

Given: $X_2 = x_2 = 1,\ E_1 = e_1 = -1, E_2 = e_2 = 1$

The Query: $P(X_2 = 1|E_1 = -1, E_2 = 1)$

$$\boldsymbol{P}(X_2|E_1 = -1, E_2 = 1) = \alpha O_2 T^T \boldsymbol{P}(X_1|E_1 = -1)$$

$$= \alpha \begin{bmatrix} 0.2 & 0 \\ 0 & 0.8 \end{bmatrix} \begin{bmatrix} 0.4 & 0.6 \\ 0.7 & 0.3 \end{bmatrix}^T \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix}$$

$$= \alpha \begin{bmatrix} 0.2 & 0 \\ 0 & 0.8 \end{bmatrix} \begin{bmatrix} 0.46 \\ 0.54 \end{bmatrix}$$

$$= \alpha \begin{bmatrix} 0.092 \\ 0.432 \end{bmatrix}$$

$$= \frac{1}{0.092 + 0.432} \begin{bmatrix} 0.092 \\ 0.432 \end{bmatrix}$$

Probability that second bit is also received correctly: $\frac{0.432}{0.524} = 0.8244$

3. **Linear and Logistic Regression** [20 marks]

a(i) Predicted value $\hat{y}$ [1 mark]

Given $\boldsymbol{w} = \begin{bmatrix} 2.0 \\ 0.3 \\ -0.1 \\ 0.05 \end{bmatrix}$ and $\boldsymbol{x} = \begin{bmatrix} 1 \\ 8 \\ 22 \\ 120 \end{bmatrix}$

Predicted score: $\hat{y} = \boldsymbol{w}^T \boldsymbol{x}$

$= 2.0 + 0.3 \times 8 - 0.1 \times 22 + 0.05 \times 120$

$= 8.2$

a(ii) Regularized Loss: L [3 marks]

Given $y = 15$, $\lambda = 0.5$, and from Q3(a), $\hat{y} = 8.2$

$$L = \frac{1}{2}(\hat{y} - y)^2 + \frac{\lambda}{2}\sum_{i=1}^{3} w_i^2$$

$$= \frac{1}{2}(8.2 - 15)^2 + \frac{0.5}{2}(0.3^2 + 0.1^2 + 0.05^2)$$

$$= 23.145625$$

a(iii) Gradients w.r.t. weights [4 marks]

$$\frac{\partial L}{\partial w_0} = (\hat{y} - y) = -6.8$$

$$\frac{\partial L}{\partial w_1} = (\hat{y} - y)x_1 + \lambda w_1 = -54.4 + 0.15 = -54.25$$

$$\frac{\partial L}{\partial w_2} = (\hat{y} - y)x_2 + \lambda w_2 = -149.6 - 0.05 = -149.65$$

$$\frac{\partial L}{\partial w_3} = (\hat{y} - y)x_3 + \lambda w_3 = -816 + 0.025 = -815.975$$

a(iv) Iteration of gradient descent. [4 marks]

$$w_i := w_i - \alpha \frac{\partial L}{\partial w_i}$$

$$w_0 = 2.0 - 0.01 \cdot (-6.8) = 2.068$$

$$w_1 = 0.3 - 0.01 \cdot (-54.25) = 0.8425$$

$$w_2 = -0.1 - 0.01 \cdot (-149.65) = 1.3965$$

$$w_3 = 0.05 - 0.01 \cdot (-815.975) = 8.20975$$

b(i) Which activation function? [2 marks]

Sigmoid as it converts the output to probabilities within the range [0,1]

b(ii) Precision and Recall values [4 marks]

$$TP = 2$$

$$FP = 1$$

$$FN = 1$$

$$TN = 1$$

$$Precision = \frac{TP}{TP+FP} = \frac{2}{3},$$

$$Recall = \frac{TP}{TP+FN} = \frac{2}{3}$$

b(iii) Precision or Recall? [2 marks]

Recall must be prioritized. Lower the threshold below 0.5 to increase recall.

4. **Deep Neural Network. [20 marks]**

a(i) $a_1^{[1]}$, $a_2^{[1]}$ and $a_1^{[2]}$ [6 marks]

$$a_1^{[1]} = \max\left(w_{11}^{[1]} * a_1^{[0]} + w_{12}^{[1]} * a_2^{[0]}, 0\right) = \max(0.1 - 0.2, 0) = 0$$

$$a_2^{[1]} = \max\left(w_{12}^{[1]} * a_1^{[0]} + w_{22}^{[1]} * a_2^{[0]}, 0\right) = \max(-0.2 + 0.1, 0) = 0$$

$$a_1^{[2]} = \max\left(w_{11}^{[2]} * a_1^{[1]} + w_{21}^{[1]} * a_2^{[1]}, 0\right) = \max(0 + 0, 0) = 0$$

a(ii) Alternative activation function. [3 marks]

The ReLU leads to zero output and it is known as dying neuron issue.

A potential solution is to use Leaky ReLU activation function.

a(iii) Three class classification. [3 marks]

The output layer contains only one perceptron. Hence it cannot be used for three class classification. To make it useful for three class classification, the output layer must contain three neurons.

b(i) Sigmoid activation function [2 marks]

Maximum value of gradient of sigmoid function is 0.25, which makes the gradients scaled down aggressively as the effect of errors are propagated back to the initial layers.

b(ii) Issue with weight initialization. [3 marks]

Initial weight values are very low, and this leads to very small initial gradients (vanishing gradient problem). Small gradients lead to very slow convergence of model parameters towards the optimal values

b(iii) Modified Neural Network [3 marks]

Yes, I agree. The revised network has skip connection and the skip connections allow gradient propagation from output even if the weights are almost zero in the subsequent layers.

5. **Convolutional Neural Networks [20 marks]**

a(i) Dimensions of Outputs of Convolution Layers. [4 marks]

Convolution Layer A	**Convolution Layer B**
32x32x128	**32x32x128**

a(ii) Identity mapping or Projection Mapping? [1 mark]

Projection mapping as input and output dimensions do not match

a(iii) Design of Skip Connection [5 marks]

The skip connection should ensure that input and output dimensions match

The input dimension: 32x32x64

The output dimension: 32x32x128

We need:

- 128 filters
- Kernel Size: 1x1
- Stride: 1
- Padding: 0

(b) Which network is parameter-efficient? [10 marks]

Aiken's Design:

# of Parameters = ($kernel\_size * kernel\_size * in\_channels * out\_channels$) + # of bias terms

Number of parameters in Aiken's design = 3*3*256*256+256=590080

Dueet's design:

# of Parameters in first 1x1 convolution layer = 1*1*256*64+64=16448

# of Parameters in first 3x3 convolution layer = 3*3*64*64+64=36928

# of Parameters in second 1x1 convolution layer = 1*1*64*256+256=16640

Number of parameters in Dueet's design = 16448+36928+16640 = 70016

Dueet's bottle neck is parameter efficient

6. **Markov Decision Processes [10 marks]**

(a) Utilities at states S_1 and S_2 with $\pi(S_1) = \pi(S_2) = right$ [4 marks]

Given: $U(S_3) = 10$.

Utilities of S_2 and S_1 can be calculated as:

$$U(S_2) = -1 + 0.9 * 10 = 8$$

$$U(S_1) = -1 + 0.9 * 8 = 6.2$$

(b) An iteration of value iteration algorithm [6 marks]

Given: $U_0(S_1) = 0$, $U_0(S_2) = 0, and\ \ U(S_3) = 10$

An iteration of Value Iteration algorithm:

State S_1:

$$U_1(S_1) = \max(-1 + 0.9 * U_0(S_1), -1 + 0.9 * U_0(S_2)) = -1$$

State S_2

$$U_1(S_2) = \max(-1 + 0.9 * U_0(S_1), -1 + 0.9 * U(S_3)) = 8$$

Rough Work

=== END OF PAPER ===